RM9-R1: What does the aggregate observer measure? Source findings and a concrete contract decision

Method-first ramp-metering review

5 October 2026

Status. The accepted source-first review has reached UG4-O. No speed map, new physical law or simulation is adopted. The new section-speed Candidate C and the historical passage Candidate A are distinct hypotheses. Physical passage realization Candidate P keeps the mapping unresolved until further analytic grounding. Detailed Pros/Cons and the decision are in the companion findings. This handout supplies the mathematical definitions.

1 Three different observation supports

Let i identify a longitudinal road cell, $\ell \in \{1, 2\}$ a mainline lane, L_i its length, h a common time step and $t_s = t_0 + sh$. Stored continuous mass $n^s_{i\ell}$ is measured in vehicle equivalents. Define road-total mass and road-total density (summed over lanes):

$$N^s_i = n^s_{i1} + n^s_{i2}, \qquad K^s_i = N^s_i / L_i. \tag{1}$$

An average density per lane would be $K^s_i/2$; the benchmark uses the summed normalization consistently. True spatial inventory is an idealized observation, not detector occupancy or density inferred from detector speed.

At a labeled face e, let $q^s_{e\ell}$ be an actual accepted longitudinal lane rate during the completed interval $[t_s, t_{s+1})$; Q^s_e is its road total and B^s_e is cumulative passage mass:

$$Q^s_e = \sum_\ell q^s_{e\ell}, \qquad B^{s+1}_e - B^s_e = hQ^s_e. \tag{2}$$

At a merge, the face label must distinguish donor-mainline passage from combined downstream passage including ramp entry. Counts and interval rates need not be independent information, but both are accounted for.

Passage time mean. For vehicles a crossing a station/window with longitudinal speeds u_a, the arithmetic passage mean is

$$V_{\text{pass}} = \frac{1}{m} \sum_{a=1}^{m} u_a. \tag{3}$$

If a lane has one representative passage speed, pooling lanes uses passage counts (equivalently constant interval rates) as weights. This is not the speed of the traffic remaining inside an upstream cell. At zero passage the statistic is unavailable, not a measured zero.

Instantaneous section mean. A density-weighted longitudinal mean over the traffic present in a section is

$$V_{\text{sec}}(t) = \frac{\sum_\ell \int_{\text{cell } i} k_\ell(x,t) u_\ell(x,t)\, dx}{\sum_\ell \int_{\text{cell } i} k_\ell(x,t)\, dx}, \qquad N_i(t) > 0. \tag{4}$$

This remains meaningful when vehicles are present but no one crosses the face. Neither this definition nor the passage definition requires an upstream-cell-to-face equality.

2 Same-region identities are not arbitrary cell/face identities

For a rectangular longitudinal space–time region $\mathcal{A}$ of area $L\Delta t$, let d_a be the longitudinal distance traveled by vehicle a inside it, and r_a its time spent inside it. The generalized definitions are

$$q_{\mathcal{A}} = \frac{\sum_a d_a}{L\Delta t}, \qquad k_{\mathcal{A}} = \frac{\sum_a r_a}{L\Delta t}, \qquad u_{\mathcal{A}} = \frac{\sum_a d_a}{\sum_a r_a} = \frac{q_{\mathcal{A}}}{k_{\mathcal{A}}}. \tag{5}$$

For multiple lanes, sum longitudinal distance and time across the lanes using the same normalization. Lateral motion itself is not extra longitudinal distance. These source definitions share the *same region*. Their flow is a traveled distance average, not necessarily a single cell's downstream accepted rate. Their density is a time–space average, not necessarily N_i^s/L_i at one instant. Therefore the identity does not justify defining independent passage speed as Q_e^s/K_i^s during a transient or receiving-limited interval.

For a section with no internal sources, conservation gives

$$N(t_b) - N(t_a) = B_{\text{in}}(t_a, t_b) - B_{\text{out}}(t_a, t_b). \tag{6}$$

It determines inventory change, not every interior traveled distance or crossing-speed distribution. Ramp entry is an additional labeled input where present. Unpassed traffic inside the section must not disappear from estimation.

Source outcome. FB0030 uses one-minute detector flow and speed to estimate density and acknowledges time/space-mean mismatch. FHWA distinguishes point passage and section means; dual loops can separately measure crossing speed. The inspected Jamshidnejad/De Schutter papers define generalized metrics and retain internal traffic across windows. The 2018 multilane procedure explicitly excludes lane changes in Section 4.2; a later trajectory approximation does not validate our adapted lane-changing CTM observer.

Evidence boundary. These are metric and estimation precedents. No source proves our exact full-history information collision, useful-action distinction or combined gradient chain. First-order observation closures below are reviewer proposals, not newly validated field measurements.

3 Candidate C: section speed under a declared first-order closure

Let $v_{i\ell}$ be free-flow speed, $w_{i\ell}$ backward-wave speed, $C_{i\ell}$ capacity and $k_{i\ell}^{\text{jam}} = M_{i\ell}/L_i$ jam density. Use the capacity-limited FD implied by these public parameters:

$$f_{i\ell}^{\text{eq}}(k) = \min\{v_{i\ell}k,\ C_{i\ell},\ w_{i\ell}(k_{i\ell}^{\text{jam}} - k)\}, \qquad U_{i\ell}^{\text{eq}}(k) = f_{i\ell}^{\text{eq}}(k)/k \quad (k > 0). \tag{7}$$

Candidate C assigns this one representative speed to each occupied lane cell at the stored-state epoch, with $k_{i\ell}^s = n_{i\ell}^s/L_i$:

$$V_i^s = \frac{\sum_\ell n_{i\ell}^s U_{i\ell}^{\text{eq}}(k_{i\ell}^s)}{N_i^s}, \quad N_i^s > 0. \tag{8}$$

An empty lane contributes zero to the numerator without evaluating $0/0$; an empty cell has an unavailable-speed flag. Occupied stationary traffic can give $V_i^s = 0$. This closure does not add a dynamic speed state or modify G2. It is a spatially homogeneous/equilibrium observation hypothesis, not a guarantee about shocks, acceleration or actual detector speeds.

4 What C does and does not hide

Equation (8) equals $\sum_\ell f_{i\ell}^{\mathrm{eq}}(k_{i\ell}^s)/K_i^s$ under the homogeneous closure. The numerator is a constitutive *within-cell equilibrium* flow sum, not the actual accepted downstream flow Q_e^s. Thus the proposed speed channel is computed from the lane state separately from accepted face counts; no statistical or algebraic independence of all observations is claimed.

On the two-lane free-flow FD subcase with distinct known speeds,

$$N_i^s V_i^s = v_{i1} n_{i1}^s + v_{i2} n_{i2}^s, \qquad n_{i1}^s = N_i^s \frac{V_i^s - v_{i2}}{v_{i1} - v_{i2}}, \quad n_{i2}^s = N_i^s - n_{i1}^s. \tag{9}$$

This identifies the *stored* split in that subcase, even if a downstream constraint caps the face flow, provided the proposed representative-speed closure still places both cell densities on their free-flow FD branches. It illustrates the information consequence of a new observation hypothesis; it is not a measured receiving-limited speed theorem, a full R2 audit or a matched-history example. Equal public lane speeds and other FD branches need their own analysis; do not select a symmetry to force a favorable conclusion.

5 Candidate A and the physical-passage alternative P

The historical idealized passage proposal uses post-lateral masses $n_{i\ell}^{L,s}$ and actual lane face flows during the completed step:

$$V_{e,\mathrm{A}}^s = \frac{\sum_\ell q_{e\ell}^s U_{i\ell}^{\mathrm{eq}}(n_{i\ell}^{L,s}/L_i)}{Q_e^s}, \qquad Q_e^s > 0. \tag{10}$$

Besides equilibrium cell speed, this explicitly assumes that the upstream post-lateral representative speed is the representative crossing speed at that face. The source review does not validate that extra approximation in congestion. Zero passage gives an unavailable statistic. A stays viable as a conditional mathematical benchmark; its operational name is not proof of physical passage realization.

Candidate P retains actual point passage speed as indispensable, but stops information testing until a physical/spatial reconstruction law is admitted. The current inventory/accepted-flow update alone does not provide that law. Initial conditions, internal queues and any added trajectory approximation must be reviewed; no microscopic state or simulator is automatically adopted.

Recommendation for discussion: C. It matches the declared cell inventory's section support and removes the extra cell-to-face assumption. It remains an explicitly idealized first-order benchmark and may expose lane contrast. If physical point-detector fidelity is required now, choose P. Neither the selected source-first route nor this recommendation adopts C/A/P.

6 Causal history and initialization

Keep the approved ordering

$$n^s \xrightarrow{\text{lateral}} n^{L,s} \xrightarrow{\text{forward/merge}} n^{s+1}. \tag{11}$$

The decision epoch t_* occurs at a stored-state epoch after all preceding steps have completed and before the next update. A future common open-loop schedule is selected then; this does not introduce repeated feedback control. With C, observe (N_i^s, V_i^s) at stored epochs $t_s \le t_*$ and counts/flows only for completed intervals ending no later than t_*. With A, passage speeds also belong only to those completed intervals. Exclude all future observations generated by any candidate schedule from the initial information set.

Denote the admitted history by

$$\mathcal{H}_* = (\mathcal{O}_{[t_{\text{init}},t_*]},\ p,\ b_{\text{public}},\ u_{\text{past}}). \tag{12}$$

Here $\mathcal{O}$ contains the declared mainline/ramp observations, p the public model parameters, b_{public} the common known forcing (including declared future forcing), and u_{past} the prior meter permissions. Every claimed latent initial state must match every initial and historical channel. The public graph specifies section lengths, face locations, boundary lane-allocation rules and downstream supplies. No lane-dependent exogenous input is kept private merely to create ambiguity.

Trivial sufficiency boundary. If the complete initial physical state x^0, all forcing b^s, parameters and previous controls are known, deterministic recursion $x^{s+1} = F_B(x^s, b^s, u^s)$ already determines the full state. A known empty initialization is such a subcase when all modeled states are empty and all structural inputs are public. Hiddenness cannot be claimed there by ignoring the recursion. If only initial aggregate observations are admitted, unobserved lane coordinates remain candidates only within their full observation-consistent set. This is an initialization contract, not permission to invent private prehistory.

7 Ramp balances and geometry requirements

Let W_r^s be external ramp waiting, R_{pre}^s traffic upstream of the meter inside finite ramp storage, and R_{post}^s traffic in any declared post-meter region. Let A_r^s be external arrival rate, b_r^s admitted entry, g_r^s actual meter release and r_r^s actual merge entry, all in mass/time. Where these compartments exist, conservation requires

$$W_r^{s+1} = W_r^s + h(A_r^s - b_r^s), \tag{13}$$

$$R_{\text{pre}}^{s+1} = R_{\text{pre}}^s + h(b_r^s - g_r^s), \tag{14}$$

$$R_{\text{post}}^{s+1} = R_{\text{post}}^s + h(g_r^s - r_r^s). \tag{15}$$

For several post-meter cells, the last line is their sum balance and internal flows cancel. The permission limits g_r^s; finite sending/storage/receiving can further restrict it. r_r^s is set by the adopted shared receiver using the downstream ramp offer, not automatically by the current permission.

8 Minimum layout/accounting declaration before R2/G3

Declare a finite two-mainline-lane cell graph, the meter location, any finite pre/post-meter ramp regions, the ramp-entry receiver and every observed section/face. Before and after the merge, donor passage and combined passage must have distinct labels. At a simple shared receiver,

$$q^s_{\text{combined}} = m^s + r^s_r \tag{16}$$

for its two inputs; a second through lane is accounted for separately in the road-total downstream face. Ordinary through-lane transfers and ramp entry must jointly respect each receiving lane's approved supply. This is a required interface declaration, not adoption of a new topology or mandatory-merge law.

If the meter and merge are truly co-located, omit nonexistent post-meter storage and specify that specialization explicitly. Do not impose co-location merely to identify release and entry. Conversely, storage/propagation cannot be silently omitted where the task includes them. Numerical cell counts are not a separate methodological gate, but the geometry must be coherent before a realized full-history/action certificate.

The complete population includes mainline external waiting where present, mainline cell mass, W_r, R_{pre} and R_{post}. All initial traffic and unserved demand remain counted. The accepted total-time objective and UG1–UG3 dynamics are unchanged. No gradient experiment was run.

9 Evidence and disposition

Established in R1	Source-backed metric/support distinctions; conditional observer proposals; causal/ramp/initialization requirements.
Still open	UG4-O observer adoption; source-grounded concrete layout; G3 feasible equal-history/useful-action evidence; G4 validated useful gradients.
Not authorized	Traffic runs, SUMO installation, new dynamics, extra lanes, noise/smoothing, successor or deferred extension execution.

Primary references inspected.

- FB0030: lane model, printed 1768–1769.
- FHWA Traffic Monitoring Guide 2022, speed and occupancy sections.
- FHWA Data Quality White Paper, Section 4.3.2.
- Jamshidnejad/De Schutter 2015, Sections I–II.
- Jamshidnejad/De Schutter 2018 author report, Sections 2.1, 4.1–4.2 and 5.

Stop. Discuss C/A/P at UG4-O. No observer is adopted by this document.